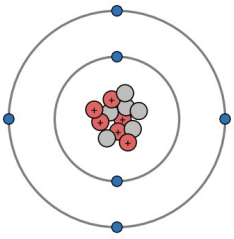
Isotopes

AQA GCSE Combined Science: Trilogy · Chemistry Topic 1 · Revision activity 2 · 16 marks · about 25 minutes

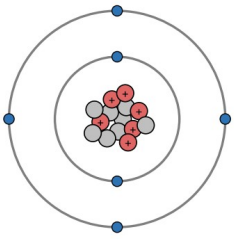
Read the key information (5 min), answer the **quick recall** questions without looking back (5 min), then answer the **exam questions** (15 min). Mark your work with the mark scheme on the last page.

Key information

- **Isotopes** are atoms of the same element with the **same number of protons** but a **different number of neutrons**.
- So isotopes have the **same atomic number** but a **different mass number**.
- Isotopes of an element have the same number of electrons, so they have the **same chemical properties** (they react in the same way).
- We name isotopes using their mass number, e.g. carbon-12 and carbon-14, or chlorine-35 and chlorine-37.



carbon-12
6 protons, 6 neutrons, 6 electrons



carbon-14
6 protons, 8 neutrons, 6 electrons

Relative atomic mass, A_r

Most elements are a mixture of isotopes. The **relative atomic mass (A_r)** is the **average** mass of an atom of the element, taking into account how much of each isotope there is (its **abundance**).

Worked example

Chlorine is 75% chlorine-35 and 25% chlorine-37.

$A_r = (35 \times 75) + (37 \times 25) \div 100 = (2625 + 925) \div 100 =$ **35.5**

Notice that the answer is closer to 35, because there is more chlorine-35.

Section A: Quick recall (6 marks – 1 mark each)

1 What is an isotope?

2 Do isotopes of the same element have the same number of protons?

3 Which particle is different in two isotopes of the same element?

4 Why do isotopes of an element react in the same way?

5 How many neutrons are in an atom of carbon-14? (atomic number 6)

6 What does A_r stand for?

Section B: Exam questions

01 Oxygen has three isotopes. The atomic number of oxygen is 8.

01.1 Copy and complete the table.

Isotope	Protons	Neutrons	Electrons
oxygen-16			
oxygen-17			
oxygen-18			

[2 marks]

01.2 Explain why these atoms are all isotopes of the same element.

[2 marks]

02 Bromine is 51% bromine-79 and 49% bromine-81. Calculate the relative atomic mass of bromine. Give your answer to 1 decimal place.

[2 marks]

03 Copper is 69% copper-63 and 31% copper-65. Calculate the relative atomic mass of copper. Give your answer to 1 decimal place.

[2 marks]

04 Lithium has two isotopes: lithium-6 and lithium-7. The relative atomic mass of lithium is 6.9. Which isotope is more common? Explain your answer.

[2 marks]

Mark Scheme – Isotopes

(1) = one mark. / separates acceptable alternatives. Words in brackets are not needed for the mark.

Question	Marking guidance	Comments	Mark
A	**1** atoms of the same element with the same number of protons but different numbers of neutrons **2** yes **3** neutrons **4** they have the same number of electrons (same electronic structure) **5** 8 **6** relative atomic mass	1 mark each	6
01.1	all protons = 8 and all electrons = 8 (1) neutrons = 8, 9, 10 (1)		2
01.2	same number of protons / same atomic number (1) different numbers of neutrons / different mass numbers (1)		2
02	(79 × 51) + (81 × 49) ÷ 100 = (4029 + 3969) ÷ 100 (1) = 79.98 = 80.0 (1)		2
03	(63 × 69) + (65 × 31) ÷ 100 = (4347 + 2015) ÷ 100 (1) = 63.62 = 63.6 (1)		2
04	lithium-7 (1) because the relative atomic mass (6.9) is much closer to 7 than to 6 (1)		2

Total: 16 marks